Project Plan: From Radiation-Induced Defects to Device Degradation in Silicon

March 20, 2026

Objective

Use Geant4 to model radiation transport in silicon and MATLAB to connect radiation-induced damage to semiconductor degradation and device performance loss.

Modules

Module 1: Geant4

Simulate proton or neutron transport in a silicon slab and extract deposited energy, depth-dependent deposition, and recoil-like events.

Module 2: Defect Model

Use a first-order relation

$$N_D = k_D \Phi$$

to relate fluence Φ to defect concentration N_D.

Module 3: Lifetime Model

Use

$$\frac{1}{\tau_{\text{eff}}} = \frac{1}{\tau_0} + K_\tau N_D$$

to represent recombination-center driven lifetime degradation.

Module 4: Device Metrics

Use diffusion length and leakage current trends:

$$L = \sqrt{D\tau_{\text{eff}}}, \qquad \Delta I = \alpha \Phi V.$$

Module 5: Cryogenic Extension

Add a temperature-dependent kinetics section for radiation-induced defects in silicon. The three first-order models are

$$D_{\text{def}}(T) = D_0 \exp\left(-\frac{E_m}{k_B T}\right), \qquad e(T) = e_0 \exp\left(-\frac{E_a}{k_B T}\right), \qquad R_{\text{ann}}(T) = R_0 \exp\left(-\frac{E_{\text{ann}}}{k_B T}\right).$$

These represent defect migration, trap emission, and annealing, respectively.

Deliverables

- Geant4 CSV outputs
- MATLAB figures for defect density, lifetime, diffusion length, and leakage current
- MATLAB cryogenic figures for migration, trap emission, emission time, and annealing versus temperature
- technical notes and slide outline